Communication System

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Notes

- Our subject code is EX 656, of which only 4 question papers are added. `Their questions are stylized as this`.
- This document encorporates EX 652 (CS-I) and EX 702 (CS-II) as well, for further enrichment of question papers.
- This may create an effect of duplication of question/year, and the authors assure that this effect, if is created, is not real.
- If a question was asked in Regular Exam, it is kept in **boldened** form. If it was asked in back exam, it is in regular font.
- The marking scheme present in the syllabus is quite weird:
  - Ch: 1, 2, 4 → 20 Marks
  - Ch: 3, 5 → 10 Marks
  - Ch: 6, 7 → 10 Marks
  - Ch: 8, 9, 10 → 20 Marks
  - Ch: 11, 12, 13 → 20 Marks
- So, it was decided the marks for individual chapters would be decided as per their ratios of course hours of the total marking scheme. Total course hours = 60.
- There are 2 2071 Magh paper (new and old back). The old back is *stylized as italic* to separate it as of a different subject code.
- Months are marked as:
  - Ba: Baisakh
  - Jth: Jestha
  - Asa: Ashar
  - Shr: Shrawan
  - Bh: Bhadra
  - Ash: Ashwin
  - Ka: Kartik
  - Mng: Mangsir
  - Po: Poush
  - Ma: Magh
  - Ch: Chaitra

Contents

1 Introduction

(2 Hours / 2.5 Marks)

1.1 Fundamental Concepts

1.1.1 A/D sources and transmission

1. Define Channel in communication system (**CS**)and classify them with example. [4+6] (73 Ma)
2. Draw the block diagram of CS and explain briefly. [6] (76 Ba, 71 Ma)
3. What major elements does a CS contains? [4] (**68 Bh**)
4. Draw the block diagram of analog CS. [2] (**72 Ash**) [4] (68 Jth)
   |→and briefly explain each block. [6] (**79 Ch**) [8] (**74 Bh**)
5. What are the main components of analog CS? Describe each. [2+5] (**69 Bh**) [4] (67 Shr)
6. Draw the block diagram of digital CS. [2] (**72 Ashh**)
7. Briefly explain the function of the basic components of a digital communication system. [5] (**79 Bh**)
8. Explain in brief the functional block diagram and the basic elements of a digital communication system. [5] (73 Shr)
9. Sketch and describe the functional digital CS working in a duplex mode and contrast it with an analog communication system. [5] (**81 Ch**)
10. Sketch a generic block diagram of a digital CS for full-duplex mode. [5] (**79 Ch**)
11. Draw functional block diagram of DCS and explain the significance of each major blocks. [6] (**80 Ch**)
12. Explain digital communication system with appropriate block diagram. [5] (80 Ba)
13. Write the advantages of digital CS over analog CS. [3] (**79 Ch**)
    |→Write the advantages of analog CS over digital CS? [4] (**77 Ch**)
14. List out the differences between analog source and digital source. [2] (**77 Ch**)
15. List the differences between digital communication system. [2] (**80 Ch**)

1.1.2 Noise, Distortion and Interference

1. Define noise. [2] (72 Ma)
2. Define any three types of noise. [3] (**76 Ch**)
3. Distinguish between external and internal noise. [2] (**75 Bh**, 72 Ma)
4. Describe the types and causes of internal and external noise that may affect the CS. [5] (**76 Bh**)
5. Describe the types and causes of any 4 types of internal noise that may affect the CS. [8] (**73 Bh**)
6. Briefly explain any four types of noise encountered in CS. [4] (68 Jth)
7. List various sources of external noise. [2] (72 Ma)
8. Explain thermal and high frequency noise. [3] (**71 Bh**)
9. How does noise limit the performance of CS? [2] (**73 Bh**)
10. What are the limitations posed by noise and interference? [3] (**70 Bh**)
11. Define noise, distortion and interference. [2] (**76 Bh**) [3] (**71 Bh**)
12. Define distortion and explain its types. [3] (81 Ba)
13. What will be the effects of noise, distortion and interference? [1] (**71 Bh**)
14. Write short notes on: Distortion and interference. [4] (71 Ma)
15. Explain briefly about linear type and non-linear type distortion. [2] (**72 Ash**)
16. List out the sources of interferences. [2] (**75 Bh**)
17. How can effects of noise and interference be minimized? [2] (75 Ba)
18. Mention its impairments and performance parameters. [3] (**81 Ch**)
19. What are the effects of atmospheric noise in analog CS. [2] (**77 Ch**)

20. Differentiate among noise, interference and distortion. [4] (70 Ma) [5] (**78 Ch**) [6] (**80 Ch**)
    |→ with examples. [6] (68 Jth)
21. Distinguish between noise and interference. [3] (**70 Bh**) [4] (75 Ba)

1.2 Introduction to Modulation

1. Define modulation. [1] (70 Ma) [2] (**69 Bh**,**71 Bh**, 67 Shr) [4] (**79 Ch**, 68 Jth)
2. "Communication over long distance is impossible without modulation. Modulation is must to mitigate several constraints in transmission." Justify. [4] (**68 Bh**)
   |→ What are the needs for modulation? [1] (70 Ma) [2] (**79 Ch**,**76 Ch**,**71 Bh**,**69 Bh**, 76 Ba, 75 Ba, 72 Ma, 68 Jth) [4] (**75 Bh**, 67 Shr)
3. Describe how channel can be modulated? [4] (72 Ma)

2 Representation of signals and systems in communication

(4 Hours / 6.5 Marks)

2.1 Review of Signals and Systems

2.1.1 Review of Signals

1. Represent Unit Step signal in terms of signal function. [4] (**79 Ch**)
2. What is signum signal? [2] (72 Ma)
3. Define band pass signal and band limited signal with example. [4] (**68 Bh**)
4. Define with examples, period and non-periodic signals. [4] (**67 Mng**)

2.1.2 Types of systems

1. Discluss linear, non-linear, casual and time invariant systems used in CS. [4] (*71 Ma*)
2. Explain how can you classify systems according to their basic properties. [4] (64 Shr)

2.1.3 Imulse Response and Transfer Function of a System

1. Define impulse response. [2] (**74 Bh**,**72 Ash**)
2. Define impulse response and transfer function of a system. [2] (**77 Ch**) [4] [**73 Bh**]
3. Explain significance of impulse response and transfer function. [2] (**73 Bh**)
4. Why is the impulse response of a system quite useful in the design of CS? [3] (**72 Ash**)
5. Prove that the output of any system is given by convolution of input and impulse response of the system. [4] (*71 Ma*) [6] (**67 Mng**)
6. Show that impulse response of an ideal low pass filter is non-casual. [4] (64 Shr) [6] (**75 Bh**)
   |→What is the impulse response of LPF? [1] (**72 Ash**)

2.1.4 LTI systems

1. Define LTI system. [2] (**71 Bh**,**70 Bh**)
2. Write the properties of LTI system. [4] (**68 Bh**)
3. What is the significance of LTI in communication engineering? [3] (**70 Bh**)
4. Find the expression (in time domain) for the output of a LTI. [5] (**71 Bh**)
5. Determine whether the following signals are: [4+4] (**81 Ch**)
   a. $y(t) = 2t^2\ x(t)\ +\ 3t\ x(t-4)$ linear or not
   b. $y(t) = x^2(t)\ +\ x(t+2)$ casual or not.
6. Determine whether the following signals are: [3+3] (**80 Ch**)
   a. $y(t) = x^2(t)$ linear or not. b. $y(t) = y(t-4) + x(t-4)$ time invarient or not.

2.2 System Transformations and Properties

2.2.1 Transformations on Transfer Function

1. State any four properties of the Fourier Transform. [4] (*71 Ma*)

2. State and explain following properties of Fourier Transform. [5] (**67 Mng**)
   a. Modulation b. Duality (symmetry) c. Time shifting
   d. Scaling e. Convolution

2.2.2 Lowpass/Bandpass signals and systems

2.2.3 Bandwidth of systems

2.2.4 Distortionless Transmission

1. Explain the significance of distortionless transmission in CS with necessary derivation. [5] (72 Ma)

2. Derive the transfer function for a system that provides distortionless transmission. [3] (**69 Bh**)
   |→and draw its amplitude response as well. [6] (**77 Ch**)

3. Explain distortionless transmission line with its frequency response. [3] (**76 Bh**, 75 Ba) [4] (76 Ba)

4. What are the conditions for distortionless transmission? Explain with necesary diagram. [6] (**74 Bh**)

5. Write short notes on distortionless transmission. [5] (**70 Bh**, 70 Ma)

2.3 Special Transforms and Signal Representations

2.3.1 Hilbert Transform and its Properties

1. On Hilbert Transform:
   |→Define. [2] (**80 Ch**,**80 Ch**,**76 Bh**,**75 Bh**, 71 Ma) [4] (73 Ma)
   |→Write short notes. [5] (**81 Bh**)
   |→Write its properties. [2] (76 Ba, 75 Ba) [3] (**76 Bh**) [4] (73 Ma)
   |→Describe, with its properties. [2] (**78 Ch**) [4] (71 Ma)
   |→Describe, with mathematical expression in frequency domain.
   [2] (**76 Bh**, 76 Ba) [3] (**80 Ch**, 75 Ba)

2. How is Hilbert Transform different from Fourier transform? [2] (73 Ma)

2.3.2 Band-pass/band-limited Signals Representation

2.3.3 Complex envelops rectangular representation

2.3.4 Polar representation

3 Spectral Analysis

(2 Hours / 2 Marks)

3.1 Foundations of Spectral Analysis

3.1.1 Spectral Analysis

1. Define energy spectral density and power spectral density function of a signal. [4] (**79 Ch**,**75 Bh**,**74 Bh**)
2. Differentiate between ESD and PSDF. [2] (70 Ma)
3. Define energy spectral density (ESD)? [2] (**71 Bh**)
4. Find ESD and total energy for sinc pulse defined by g(t) = A sinc (2wt). [3+2] (**71 Bh**)

3.1.2 Reviews of Fourier Series and Transform

3.2 Signal Energy and Power

3.2.1 Energy and Power

1. Define energy and power signal. [2] (**76 Bh**,**73 Bh**, 65 Ka)
2. Define bandwidth of a system along with necesary diagram. [2] (**73 Bh**)
3. Write properties for energy signal. [3] (76 Ba)
4. State Rayliegh energy theorem. [2] (75 Ba) [4] (72 Ma)
5. Prove Rayliegh energy theorem for a given energy signal x(t). [6] (75 Ba)
   |→Derive the expression for the Rayliegh Energy Theorem. [4] (*71 Ma*)
6. How can you calculate the power of a periodic signal when its Fourier-series coefficients C_n are known? [2] (72 Ma)
7. Determine whether the given signal $x(t) = e^{-2t}\ u(t)$ is a power or an energy signal. [5] (**80 Ch**)
8. Find whether the signal $x(t) = A\cos(2\pi ft)$ is energy or power type signal. [5] (**76 Bh**) [6] (65 Ka)
9. Determine whether a Unit Step signal is energy or power type or neither of the two. [4] (**79 Ch**)
10. Prove that for an energy signal x(t) thhe AC function and energy spectral density form Fourier transform pair. [4] (76 Ba)
11. Compute the energy and power of unit step signal. [5] (**78 Ch**)

3.2.2 Parseval's Theorem

1. State Parseval's Theorem. [2] (**80 Ch**) [3] (**77 Ch**)

3.3 Power spectral density function

1. What is PSD? [2] (**73 Bh**)

2. What is PSDF? [4] (73 Ma)

3. Define power and energy spectral density function. [2] (**72 Ash**) [4] (**69 Bh**, 67 Shr)

4. Derive an expression for PSDF of an arbitrary signal X(t). [4] (**73 Bh**) [6] (73 Ma, 70 Ma)

5. Brifly explain the operation of analog spectrum analyzer. [6] (**79 Ch**, 67 Shr)
   |→with block diagrams. [8] (65 Ka)

6. Find power spectral density and average power for the periodic signal defined by g(t) = A $\cos(2\pi f_c t + \theta)$. [6] (**69 Bh**)

3.4 AC function, relationship between psdf and AC function

1. Define auto-correlation function. [1] (76 Ba) [3] (**74 Ch**)
   |→Write short notes on: Autocorrelation function and its properties. [5] (**79 Ch**)

2. List out any three properties of AC function. [3] (**78 Ch**,**70 Bh**)

3. What is white noise. [3] (71 Ma)

4. Write short notes on: White noise and its psdf. [2] (**70 Ch**)

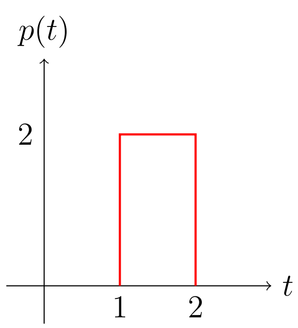
5. Mention the AC function of white noise. [3] (**78 Ch**,**70 Bh**)
   |→Explain the relation between PSDF and AC function with the example of white noise.
   [3] (71 Ma) [4] (**74 Bh**,**73 Bh**) [6] (**80 Ch**)

6. Derive the AC function of white noise utilizing PSD with necesary diagram. [4] (**75 Bh**)

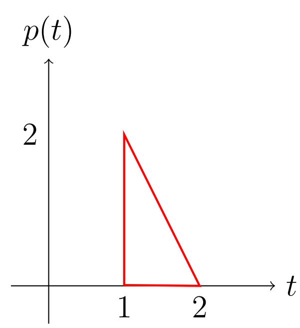
7. Prove that in LTI system when a power signal x(t) is applied the PSD of its output is equal to the PSD of its input multiplied by the squared amplitude response of the system. [5] (**77 Ch**)

8. Given a signal x(t) = $\cos(200\pi t)$, find the auto correlation of x(t) at $\tau = \frac{\pi}{4}$. [3] (**72 Ash**)

9. Derive auto-correlation, cross-correlation and convolution of the following two signals graphically:



a.



b.

[3+3] (**81 Ch**)

4 Amplitude Modulation and Demodulation

(8 Hours / 11 Marks)

4.1 Fundamentals of Amplitude Modulation

4.1.1 Amplitude Modulated (AM) signals

1. What is the basic characteristics of AM modulation. [3] (**72 Ash**)

4.1.2 Time domain expressions

4.1.3 Frequency domain representation

1. Find the frequency domain expression for standard AM wave for single tone message signal. [3] (**71 Bh**)

4.1.4 Modulation index

1. Define BW and modulation indices. [1] (**79 Ch**)
2. Draw a neat diagram of amplitude-modulated wave and drive an expression for modulation index. [6] (**72 Ash**)
3. Justify why AM transmitters are generally operated with the modulation index as close to 100% as possible. [2] (**72 Ash**)
4. Define modulation index of AM wave. [1] (**76 Bh**)
5. Short notes on: Modulation index. [4] (72 Ma)

4.1.5 Signal Bandwidth

1. Define BW. [1] (**79 Ch**)

4.2 AM System Characteristics

4.2.1 Single Tone message

4.2.2 Carrier and Side-band components

4.2.3 Powers in carrier and Side-band components

4.2.4 Bandwidth and power efficiency

1. Why is conventional AM wasteful of power and bandwidth? [4] (**73 Bh**)
2. Compare various types of AM systems in terms of transmission power and transmission bandwidth. [2] (70 Ma)

4.3 Generation of AM Variants

4.3.1 Generation of DSB-FC AM (Balanced, Ring Modulators)

1. Derive the expression for double tone AM. [3] (**80 Ch**) [6] (**79 Ch**)

2. Find the time domain and frequency domain expressions for signle tone DSB-FC AM modulated wave. Also, show the spectrum of the modulated signal. [3+3+3] (75 Ba)

3. Explain the generation of DSB FC signal using square law modulator with appropriate diagram and spectrum of transmitted signal. [5] (**76 Bh**) [6] (**80 Ch**)

4. Describe the process of generating DSB-SC AM using Balance Modulator. [5] (**75 Bh**)

5. Explain Ring Modulator method of generating DSB-SC signal. [5] (**70 Bh**)
   |→with waveforms and necesary derivation. [6] (73 Ma) [8] (**74 Bh**)

6. Explain the method of conventional AM generation by using switching modulator. [6] (**73 Bh**)

7. Explain the generation of DSB-FC AM using switching modulator with the help of diagram and expressions. [8] (71 Ma)

8. Explain any one method generating DSB-FC AM. [4] (70 Ma)

4.3.2 DSB-SC AM: time & frequency domain, powers, bandwidth, efficiency

1. Show that a square-law modulator can be used to generate DSBAM signal with explanation of its diagram and also sketch spectrum at the input of Bpf. [5] (**72 Ash**)

2. What are the advantages of SSB-AM over DSBFC-AM? [2] (71 Ma)

3. (Assumed) Find the detection gain for SSB-SC demodulation and compare with DSB-SC. [8] (**75 Ch**)

4. (Assumed) Calculate detection gain for DSB-FC, DSB-SC and SSB and also compare them. [8] (**73 Ch**)

5. (Assumed) What is detecting gain. Prove that for 100% modulation of (DSB-AM), the detection gain is less than 1. [8] (**74 Ch**)

4.3.3 Single Side Band (SSB) Modulation: time and frequency domain, powers

1. How is DSB different from SSB signal? [2] (**79 Ch**)
   |→Compare DSB-AM and SSB wave in terms of transmission power and bandwidth. [4] (73 Ma)
   |→Compare the basics of DSB-AM, DSB-SC and SSB modulations with their respective spectrum. [5] (**72 Ash**)
   |→Compare DSB-AM, DSB-SC and SSB in temrs of complexity, power and bandwidth efficiency. [8] (*71 Ma*)
   |→How deos SSB differ from conventional AM and DSB-SC? [3] (**75 Bh**)
   |→How is SSB different from conventional full carrier AM? [2] (**70 Bh**)

2. Derive the equation for SSB modulated signal in terms of Hilber Transformation of modulating signal mt(t). [6] (**79 Ch**)
   |→Derive the expression for SSB signal. [4] (71 Ma)

3. Write application areas of SSB communication. [2] (76 Ba)

4.3.4 Generation of SSB (SSB filters and indirect method)

1. Briefly discuss any one method of generating SSB signal. [4] (**79 Ch**)

2. Describe the process of generation of SSB-AM wave using phase discrimination method. [6] (**76 Bh**)

3. Explain the phase shift method of generation of SSB AM modulated wave. [6] (75 Ba)
   |→With the help of block diagram and expression explain the phase shift method for generation of SSB-AM wave. [6] (**71 Bh**)

4. Derive the expression for SSB wave modulated by a low pass signal m(t). [6] (70 Ma)

5. What are the pros and cons of phase shift method? [2] (75 Ba)

4.4 Advanced AM Techniques

4.4.1 Introduction to Vestigial Side Bands (VSB)

1. What is vestigial side band modulation? [2] (**70 Bh**)

2. What is the motivation behind using VSB? [2] (**70 Bh**)

3. Explain about generation of VSB AM with its frequency spectrum. [4] (76 Ba)

4. Why is VSB suitable for television transmission? [2] (**70 Bh**, 76 Ba)

4.4.2 Introduction to Independent Side Bands (ISB)

4.4.3 Introduction to Quadrature Amplitude Modulations (QAM)

4.5 Demodulation of AM Signals

4.5.1 Demodulation of DSB-FC, DSB-SC and SSB using synchronous detection

1. Compare and constrast DSB-SC and SSB. [3] (**80 Ch**)

2. Explain the synchronous demodulation for DSB-SC signal along with its requirements and limitations. [6+3] (**80 Ch**)
   |→How can synchronous demodulator be used to detect DSB-SC wave? [3] (**70 Bh**)

3. Explain mathematically, the effects of phase error & frequency error in local oscillator while demodulating DSB-SC. [3] (**70 Ch**)

4. Why DSB-SC detection is known as synchronous detection, explain with required details. [3] (**72 Ash**)

5. Propose one of practically synchronised receiving system for DSB-SC wave. [5] (**76 Bh**)

6. Compare the performance of DSB-AM, DSB-SC, SSB-AM VSB. [2] (**76 Bh**)

4.5.2 Square law and envelope detection of DSB-FC

1. What is the limitation of square law detector for DSB-AM detection? [2] (70 Ma)

2. Write short notes on: Envelope Detector. [4] (**76 Bh**)
   |→Explain envelope detector. Include in your explanation how the values of capacitor and resistor can be chosen so that the output of detector is the envelope of the signal as its input. [6] (**70 Bh**)

3. Explain the process of demodulation of AM wave using envelope detector. [6] (76 Ba)
   |→with necessary conditions for time constants and waveforms. [4+2+2] (75 Ba)
   |→Draw the circuit diagram and the waveforms and describe how envelope detector can be used for demodulation of standard AM wave. [5] (**71 Bh**) [6] (70 Ma)

4. Does envelope detector demodulate dSB-SC AM wave? Explain. [2] (76 Ba)

4.5.3 Demodulation of SSB using carrier reinsertion

1. Describe any one method of demodulating DSB-FC AM signal. [8] (**75 Bh**)

4.5.4 Carrier recovery circuits

4.6 Phase Locked Loop (PLL) in AM

4.6.1 PLL: basic concepts, definition, equations and applications

1. Write short notes on: Phase Locked Loop. [5] (**79 Ch**)

2. What are the essential components that constitute PLL? [2] (**74 Bh**)

3. How AM can be demodulated using PLL circuit? Explain. [5] (**72 Ash**)

4.6.2 Demodulation of AM using PLL

1. How can PLL be used to demodulate AM signal? [6] (**74 Bh**,**71 Bh**)
   |→Explain the demodulation of AM using PLL. [4] (71 Ma)

2. (Assumed) Explain the effect of phase and frequency error in local oscillator in demoudlating DSB and SSB using PPL. [8] (**80 Ch**)
   |→Evaluate the effect of small phase error in the local oscillator on synchronous detection of DSB-SC AM. [3] (**76 Bh**)
   |→Show the effect of phase error in coherent detection of DSB-SC AM. [4] (71 Ma)

3. With block diagram and nceccessary mathematics, show that the Costas loop can be used as a practical syncrhonous receiving system suitable for use with the DSB-SC modulated wave. [10] (73 Ma)
   |→Draw the block diagram of Costas Loop detector and explain how it demodulates DSB-SC AM and corrects for phase error. [4+4+2] (**73 Bh**)

4.7 Numericals

1. For the given modulated signal draw line spectrums. [3] (**80 Ch**)
   $s(t) = 50\cos(2\pi\, 10^6 t + 60^0) - 15\ \cos(2\pi\, 10^6 t - 60^0) + 20\sin(4\pi\, 10^4 t - 190^0) + \sin(2\pi\, 10^4 t + 190^0)$.

2. For an amplitude modulated signal. [2+2+2] (**80 Ch**)
$s(t) = 50\ \cos(2\pi\ 10^6\ t) + 20\ \cos(2\pi\ 10^6\ t)\ \cos(2\pi\ 10^3\ t) + 12\ \cos(2\pi\ 10^6\ t)\ \cos(4\pi\ 10^2\ t)$
   a. Calculate the net modulation index.
   b. Calculate the total modulated power.
   c. Draw the spectrum of the signal.

3. A message signal m(t) = $10\cos(2\pi\ 4000t)$ Volt is used to frequency modulate the carrier signal c(t) = $80\cos(2\pi\ 10^7 t)$ Volt. Assuming the frequency sensitivity of the frequency modulator to be 400 Hz/Volt, calculate: [4x2.5] (**79 Ch**)
   a. Peak frequency deviation
   b. Modulation index
   c. Bandwidth of modulated signal for over 98% of FM power
   d. Total modulated signal power dissipated in unit impedance

4. An amplitude modulated signal is represented by [1+2+1+2+1] (**76 Bh**)
S_{AM}(t) = $10(1\ +\ 0.2\cos(2\Pi\ 10^3\ t))\ \cos(2\Pi\ 10^6\ t)$.
   a. Identify which type of modulation.
   b. Find modulating frequency and carrier frequency
   c. Bandwidth of the signal
   d. Carrier power, total power, power in side bands
   e. Efficiency

5. The signal m(t) = $10\cos(2\cdot 10^3\pi t)\ +\ 15\cos(1\cdot 10^3\pi t)$, c(t) = $20\cos(3\cdot 10^6\pi t)$ are applied to AM. Find [2x4] (76 Ba)
   a. The equation of the resulting signal
   b. Modulation index
   c. Total power
   d. Draw spectrum

6. An amplitude modulated wave is given by: [3+2+2+1] (**75 Bh**)
s(t) = $100\cos(2\pi\cdot 10^6 t)\ +\ 30\cos(2\pi\cdot 10^6 t)\ \cos(2\pi\ 10^3 t)\ +\ 40\cos(2\pi\cdot 10^6 t)\ \cos(4\pi\cdot 10^2 t)$ Volt
   a. Draw the frequency spectrum of modulated wave
   b. Net modulation index
   c. Total modulated power
   d. Efficiency

7. A cosine carrier of frequency 750 kHz is amplitude modulated by another cosine wave of frequency 325 Hz resulting in maximum and minimum carrier amplitudes of 110V and 90V respectively.
   a. Draw the waveform of AM wave thus created. [2+3+2] (75 Ba)
   b. Write the expression of the resulting AM wave.
   c. Find the total radiated and efficiency.

8. An AM wave is represented by s(t) = $5\,[1\ +\ \cos(6280t)]\cos(2\pi\ 10^4 t)$ volts, then find the followings: (a) Modulation Depth, (b) Maximum and Minimum Amplitude of AM wave, (c) Frequency components in modulated signal and their amplitudes and (d) Power dissipated across 1 kΩ resistor. [2+2+2+2] (**74 Bh**)

9. An amplitude modulated wave is given by: [4+3+3] (**73 Bh**)
s(t) = $50(1 + 0.3\cos(3141.60t) + 0.2\cos(2513.28t))cos(10^6 t)$
   a. Draw the amplitude spectrum of s(t)
   b. Determine the bandwidth of s(t)
   c. Calculate the power efficiency

10. An audio signal given as $15\sin(2\pi \cdot 1500t)$ amplitude modulates a carrier given as $60\sin(2\pi \cdot 100,000t)$ determine the following: [2+2+2+2] (72 Ma, *71 Ma*)
    a. Construct the modulated wave
    b. Determine the modulation index and percent modulation
    c. What frequencies would present in a spectrum analysis of the modulated wave?
    d. Sketch audio and carrier wave

11. The signals m(t) $= 5\cos(2 \cdot 10^3\pi t) + 10\cos(1 \cdot 10^3\pi t)$, c(t) $= 15\cos(2 \cdot 10^6\pi t)$ are applied to AM. [2+2+3] (**72 Ash**)
    Find modulation index
    Find total power
    Draw spectrum

12. A certain AM transmitter radiates 10 kW with the carrier unmodulated and 11.8 kW when the carrier is sinusoidal modulated. Calculate the modulation index. If another sine wave, corresponding to 30 percent modulation, is transmitted simultaneously, determine the total radiated power. [3+4] (**72 Ash**)

13. An AM wave is represented by S_{AM} (t) $= 20(1 + 0.8\cos(2\pi 1000t))\cos(9424777.96t)$ volts. Find [2x4] (**71 Bh**)
    a. Amplitude of all frequency components
    b. Modulation index
    c. Maximum and minimum amplitude of AM wave
    d. Frequency of USB and LSB

14. Given the modulated wave, u(t) $= [20 + 2\cos(3000\pi t) + 10\cos(6000\pi t)]\cos(2\pi f_c t)$ where $f_c = 10^5$ Hz. [2x4] (70 Ma)
    a. Sketch the spectrum of the signal.
    b. Find the power contained in each frequency component.
    c. Calculate efficiency.
    d. Transmission bandwidth of the system.

15. The amplitued modulated signal is given by x(t) $= 50\cos(2\pi \cdot 10^6 t) + 15\cos(2\pi \cdot 10^6 t)\ \cos(2\pi \cdot 10^3 t) + 20\cos(2\pi \cdot 10^6 t)\ \cos(4\pi \cdot 10^2 t)$. [3+2+2] (**70 Bh**)
    a. Draw the spectrum.
    b. Find the total modulated power.
    c. Find the net modulation index.

5 Angle Modulation and Demodulation

(6 Hours / 8 Marks)

5.1 Fundamentals of Angle Modulation

1. What is angle modulation? [2] (75 Ba) [4] (73 Ma)

5.1.1 Basic definitions for FM and PM

5.1.2 Time domain expressions for FM and PM

1. Derive the general expression for frequency modulation. [3] (*71 Ma*)

5.2 Spectral Analysis of FM

5.2.1 Time domain expression for single tone modulated FM

1. Find the time domain and frequency domain expression of single tone modulated FM signal. [3+3] (75 Ba)

5.2.2 Spectral representation using Bessel's function

5.2.3 Bessel's function properties

1. What are the properties of Bessel function? [2] (**76 Bh**)
2. Derive expression for single tone modulated FM signal in terms of Bessel coefficients. [8] (**75 Bh**)
   |→Find the time domain expression for single tone FM modulated waves in terms of Bessel coefficients. [6] (**71 Bh**)
3. Derive the time domain expression of single tone FM in terms of Bessel's function $J_n(\beta)$. Use the result to find the expression of average power of FM signal whose significant Bessel's coefficients are taken from '-n' to '+n'. [6+2] (**74 Bh**)
4. Express angle modulated wave in terms of Bessel function $j_n(\beta)$ of first kind of order n and argument β. [6] (**72 Ash**)

5.2.4 Bandwidth of FM: Carson's rule

1. Explain Carson's rule for determining the bandwidth of an FM wave. [2] (72 Ma)
2. Under what condition the bandwidth of FM signal is same as that of AM? Explain. [4] (72 Ma)

5.2.5 Narrow and wideband FM

1. Define narrow band and wide band FM. [2] (76 Ba)
2. How is the spectrum of Narrow band FM similar to and different from the spectrum of conventional AM? [5] (**70 Bh**)
3. Find the time domain expression for Narrowband FM signal. [6] (**73 Bh**)
4. How NBFM can be used to generate Wideband FM signal. [4] (**73 Bh**)
5. Write short notes on: Threshold effect in WBFM. [4] (**80 Bh**) [5] (**75 Bh**)

5.3 Generation and Demodulation

5.3.1 Generation of FM (direct and Armstrong's method)

1. Explain the Armstrong's method for Frequency Modulation. [4] (**80 Ch**) [6] (73 Ma)
   |→Explain how NBFM is generated by Armstrong's method. [5] (**70 Bh**)

2. Explain the process of generation of wide band FM wave using Armstrong method. [6] (76 Ba)

3. Explain with block diagram how can you generate FM using phase modulator. [5] (*71 Ma*)

5.3.2 Demodulation of FM and PM signals

5.3.3 Synchronous (PLL) and non-synchronous (limiter-discriminator) demodulation

1. What is the role of amplitude limiter in limiter discriminator method? [2] (**76 Bh**, 70 Ma)

2. Explain demodulation of FM using limiter-discriminator method. [6] (71 Ma) [7] (**71 Bh**)

3. Explain FM demodulator using PLL. [4] (**76 Bh**, 72 Ma)

4. Define lock, range, capture range in PLL. [2] (72 Ma)

5.4 FM Systems and Applications

5.4.1 Stereo FM: spectral details, encoder and decoder

1. Explain the stereo FM encoder and decoder with spectral diagram. [7] (**78 Ch**)
   |→With functional block diagrams and spectral details explain the operation of stereo encoder and decoder. [5+5] (**79 Ch**)

2. Draw the block diagram of stereo FM encoder and decoder. Explain each block briefly. [8] (76 Ba)
   |→Explain the functional block diagram of stereo encoder. [4] (70 Ma) [5] (**75 Bh**)

3. Write short notes on: Stereo encoder. [4] (71 Ma) [5] (**80 Ch**, 73 Ma)

5.4.2 Pre-emphasis and de-emphasis networks

1. Write short notes on: Pre-Emphasis and De-Emphasis. [4] (**74 Bh**) [5] (**81 Ch**,**80 Ch**)

5.4.3 The super-heterodyne radio receivers for AM/FM

1. Explain the operation of superheterodyne radio receiver. [5] (**70 Bh**) [8] (**80 Ch**, 71 Ma)

2. Write short notes on: Super heterodyne Receiver. [4] (**74 Bh**, 76 Ba) [5] (70 Ma)

3. Write short notes on: AM radio receiver. [4] (72 Ma)
   |→Write short ntoes on: FM radio receiver. [4] (**72 Ash**)

4. What are the requirements for a good radio receiver? [3] (**70 Bh**)

5. List the major differences between AM and FM superheterodyne receiver. [3] (72 Ma)

6. Why are superheterodyne receivers provided with automatic gain control (AGC) mechanism? [2] (72 Ma)

5.5 Numericals

1. The equation of angle modulated signal is v(t) = 12 $\sin(10^6 t + 5\sin(10^4 t))$. Determine the following: [4x 1.5] (**80 Ch**)
   a. Carrier frequency
   b. Modulating frequency
   c. Modulation index
   d. Power dissipate in 100Ω resistors

2. The angle modulated signal is given by S(t) = $20\cos(6 \cdot 10^8 t + 7\sin(1250t))$.
   Determine: [2.5x4] (73 Ma)
   a. The carrier and modulating frequency
   b. The modulation index
   c. Maximum frequency deviation
   d. Power dissipated in 10 Ω resistor

3. Show that a FM signal consists infinite number of cosine signal components centered at frequencies f_c + n f_m, where f_c = carrier frequency, f_m = message frequency and n = 0, ±1, ±2, ±3, ... [6] (**76 Bh**)

4. A sinusoidal modulating signal m(t) = $5\cos(18849.55t)$ is applied to an FM modulator that has frequency sensitivity of 9 kHz/V. The amplitude of carrier is 25V and the frequency is 88.7 MHz. Compute: (a) Modulation index (b) Carrier frequency swing (c) Bandwidth using Carson rule (d) Total power delivered to 10Ω resistor. [2+2+2+2] (**76 Bh**, 76 Ba)
   |→Compute: (i) Peak frequency deviation, (ii) Modulation index, (iii) Frequency swing, (iv) Carson's bandwidth. [2x4] (**71 Bh**)

5. A 102.4 MHz carrier signal is frequency modulated by 5 kHz sine wave. The resultant FM signal has frequency deviation of 75 kHz. Now, determine the followings: (a) carrier swing of FM signal, (b) the bandwidth occupied by FM signal and (c) modulation index. [3+3+3] (75 Ba)

6. An Armstrong FM modulator is required in order to transmit an audio signal of bandwidth 50 Hz to 15 kHz. The Narrow Band (NB) phase modulator used utilizes an oscillator providing carrier frequency f_{c_1} = 0.2 MHz. The output of the NB phase modulator is multiplied by n_1 by multiplier and then passed to mixer with a local oscillator frequency f_c = 10.925 MHz. The desired FM wave at the transmitter output has a carrier frequency f_c = 90 MHz and a frequency deviation deviation Δf = 75 kHz, which is obtained by multiplying the mixer output frequency with n_2 by using another multiplier. Find n_1 and n_2. Assume that the NBFM signal at the output of NB phase modulator has modulation index, $\beta = 0.5$. [9] (**74 Bh**)

7. Generate an FM signal with F_c = 50 MHz and $\beta = 1$ from a NBFM signal with F_c = 1 MHz, β = 0.1 and Fm = 10 kHz. [6] (72 Ma)

8. A carrier frequency 10^6 Hz and amplitude 3 volts is frequency modulated by a sinusoidal modulating signal frequency 500 Hz and peak amplitude 1 volt. The frequency deviation is 1 kHz. The level of the modulating waveform is changed to 5V peak and the modulating frequency is changed to 2 kHz. Write the expression for the new modulated waveform. [6] (**72 Ash**)

9. The equation of an angle modulation voltage is E = $10\sin(10^8 t + 3\sin(10^4 t))$. Calculate the carrier and modulating frequency, modulating index and power dissipated in 100 Ω resistor. [2+2+2+2] (71 Ma)

10. In an FM system a baseband signal band limited to 10 kHz modulates 100 MHz carrier wave so that the frequency deviation is 75 kHz. Find: [4+4] (70 Ma)
    a. Carrier frequency swing in the FM signal and modulation index
    b. The practical bandwidth of the FM signal

11. A message signal x(t) = $\cos(5000\pi t)$ is quantized in 128 levels using Nyquist sampling rate: [2+3+2] (71 Shr)
  a. Find SQNR of the PCM signal
  b. Find the sampling frequency required when same signal uses delta modulation for same SQNR.
  c. If the system uses DM using Nyquist sampling rate, find SQNR degradation in DM as compared to PCM.

6 Source Coding and Sampling Theory

(4 Hours / 4 Marks)

6.1 Digital Communication Framework

6.1.1 Digital communication sources, transmitters, channels and receivers

6.2 Source Coding

6.2.1 Source coding and coding efficiency

1. What is source coding? [1] (70 Asa)
2. What is the aim of source coding? [2] (**79 Ch**)
3. Discuss the importance of source coding in digital communication system. [2] (**71 Ch**,**70 Ch**, 71 Shr) [4] (**75 Ch**) [6] (**73 Ch**)
4. Distinguish between the Source coding and Channel coding. [3] (74 Ash)
5. How does source encoding differ from channel encoding? [4] (**81 Bh**)

6.2.2 Shannon-Fano and Huffman codes

1. Explain Shannon-Fano coding. [3] (73 Shr)
2. Explain Huffman codes with examples. [4] (**72 Ch**)
3. Write algorithm for Huffman's coding. [3] (70 Asa)
4. Encode "Engineer Ko Betha Arulai Ke Thaha" using Huffman Encoder and calculate the transmission efficiency. Find the codeword if the message is received as "Engineer Ko Betha Ke Ke". [7+2+3] (**81 Ch**)
5. Encode the message "Mississipi is missing" using weighted Huffman code and calculate transmission efficiency. [8] (**80 Ch**)
6. Encode "Kun Mandir Ma Janchhau Yatri" using Huffman codes and finds its efficiency. [8] (**79 Ch**)
7. A DMS has five symbols with probability of $p(x_1) = 0.4$, $p(x_2) = 0.19$, $p(x_3) = 0.16$, $p(x_4) = 0.15$ and $p(x_5) = 0.1$. Construct the Huffman code for X and calculate its efficiency and compare with fixed Length coding. [4] (81 Ba)
8. A DMS has size symbols with probability of $p(x_1) = 0.3$, $p(x_2) = 0.2$, $p(x_3) = 0.25$, $p(x_4) = 0.12$, $p(x_5) = 0.08$ and $p(x_6) = 0.05$. Construct Shannon - Fano code for X and calculate its efficiency and code Redundancy. [5] (80 Ba)
9. A discrete memoryless source has an alphabet of seven symbols whose probabilities occurrence are as under:

Symbol	S_0	S_1	S_2	S_3	S_4	S_5	S_6
Probability	0.25	0.25	0.125	0.125	0.125	0.0625	0.0625

Determine the Huffman code for above source. [5] (**79 Bh**)

10. Calculate coding efficiency of a six symbol source with probabilities P = 0.36, 0.18, 0.12, 0.09, 0.07 using Shannon Fano and Huffman's coding techniques. [8] (**76 Ch**)

11. Compute coding efficiency of a source with symbols { A_0, A_1, A_2, A_3, A_4} with corresponding probabilities 0.4, 0.3, 0.15, 0.1, 0.05 using (i) Binary coding (ii) Shannon-Fano Coding (iii) Binary Huffman Coding. [2+2+2] (76 Ash)

12. A DMS emits one of the seven symbol with probabilities P = [0.25, 0.2, 0.15, 0.08, 0.07, 0.03, 0.02]. Find the coding efficiency, code redundancy for both Shannon-Fano coding and Fixed length coding and compare result. [6] (**75 Ch**)

13. A discrete memory less source emits one of the eight symbols with probabilities P = { 0.25, 0.2, 0.15, 0.08, 0.07, 0.03, 0.02 }. If the output symbols are encoded using Huffman code, find the coding efficiency and output bit if the symbol rate of the source is 1000 symbols per second. [4+2] (71 Shr)

14. Six messages are transmitted with probabilities 0.3, 0.08, 0.1, 0.15, 0.25 and 0.12 respectively. Obtain their respective shannon-fano and Huffamn's codes and code efficiencies. [8] (75 Ash)

15. A discrete memoryless source has an alphabet of five symbols S_0, S_1, S_2, S_3, S_4 with probabilities of 0.55, 0.15, 0.15, 0.1 and 0.05 respectively. Determine the Huffman code for each symbol and calculate high the coding efficiency. [2+2] (74 Ash)

16. The source of information symbols { A0, A1, A2, A3 and A4} have corresponding probabilities { 0.4, 0.3, 0.15, 0.1 and 0.05}. Encode the source symbols using Huffman encoder and Shannon-Fano encoder and compare their efficiency. [8] (**74 Ch**)

17. A discrete memory less source emits five symbols with probabilities P = 0.3, 0.25, 0.2, 0.15, 0.1, find coding efficiency for both fixed and shannon-fano coding and compare results. [4] (**73 Ch**)

18. If a source emits symbols X_i = { A, B, C, D, E, F } in the BCD format with probabilities $P(X_i)$ = { 0.3, 0.1, 0.02, 0.15, 0.4, 0.03 } at a rate R_5 = 14.4 kBaud, find the following: [5] (**72 Ch**)
    a. Information rate
    b. Coding efficiency both with BCD and Huffman coded signal

19. A discrete memory less source emits four symbols with probabilities P = { 0.125, 0.125, 0.25, 0.5 }. If the output symbols are encoded using Shannon-Fano code, find the coding efficiency and compare the coding efficiency with that of BCD code. [4+2] (**71 Ch**)

20. Develop Huffman coding of a 5 symbol source with probabilities: $S_0 = 0.3$, $S_1 = 0.25$, $S_2 = 0.2$, $S_3 = 0.15$, $S_4 = 0.1$. And also calculating coding efficiency. [3+1] (70 Asa)

6.2.3 Coding of continuous time signals (A/D conversion)

6.3 Sampling Theory

1. Define sampling. [1] (76 Ash)

2. State and prove sampling theorem. [5] (73 Shr)

3. Explain the types of sampling techniques with waveforms. [6] (**75 Ch**)

4. State sampling theory in terms of transmitter and receiver. [4] (74 Ash)

6.3.1 Nyquist-Kotelnikov sampling theorem

1. State and prove Nyquist sampling theorem. [8] (**80 Bh**)

2. State and explain Nyquist sampling theoremm. [4] (80 Ba)
   |→State and explain Nyquist-Kotelnikov sampling theorem with time domain and frequency domain analysis. [6] (71 Shr)

3. State Nyquist sampling theory. [2] (**72 Ch**,**71 Ch**)

4. Find the Nyquist rate and the Nyquist interval for the signal x(t) = $\sin(800\pi t)/\pi t + 10\cos(2000t)$. [5] (**79 Bh**)

5. Twenty four voice signals are sampled uniformly and then have to be time division multiplexed. The highest frequency component for each voice signal is equal to 3.4 kHz. Now, [4+3] (**76 Ch**)
   a. If the signals are pulse amplitude modulated using Nyquist rate of sampling, what would be the minimum channel bandwidth required.
   If the signal are pulse code modulated with an 8 bit encoder, what would be the sampling rate? The bit rate of the system is given as 1.5 x 10^6 bits/sec.

6. An analog baseband signal, band limited to 100 Hz, is sampled at the Nyquist rate. The samples are quantized into four emssage symbols that occur independently with probabilities p1 = p4 = 0.125 and p2 = p3. Determine the information rate (bits/sec) of the message source signal. [6] (76 Ash)

7. Determine the Nyquist rate and Nyquist interval for a continuous time signal x(t) = $6\cos(50\pi t) + 20\sin(300\pi t) - 10\cos(100\pi t)$. It is to be sampled and quantize using 512 levels. [5] (**72 Ch**,**71 Ch**)

8. With mathematical derivation show that original band limited signal band limited signal can be reconstructed from its samples taken at Nyquist rate. [5] (70 Asa)

6.3.2 Time domain and frequency domain analysis

6.3.3 Spectrum of sampled signal

6.3.4 Reconstruction of sampled signal

6.4 Practical Sampling Considerations

6.4.1 Ideal, flat-top and natural sampling processes

1. Explain the aperture effect during flat-topped sampling. [2] (**78 Ch**)

2. Explain flat top sampling with appropriate derivation. [6] (80 Ba)
   |→Explain flat top sampling and compare its advantage over natural sampling. [5] (**79 Bh**)

3. State the practical considerations that need to be considered during sampling. Explain natural sampling with appropriate derivation. [3] (**81 Bh**)

4. Explain natural sampling with appropriate derivation. [6] (**81 Bh**)

5. Explain the merits and demerits of flat-top sampling of a continuous time signal using relevant mathematical expressions. [7] (81 Ba)

6. Explain with proper illustration the Natural Sampling and Flat Top Sampling. [5] (76 Ash)

7. Illustrate and explain the ideal sampling and reconstruction of sampled signal. [6] (75 Ash)

8. What are the practical factors to be considered while sampling? [6] (**70 Ch**)

9. If two band limited signals $X_1[t]$ and $X_2[t]$ have bandwidths of W_1 and W_2 Hertz respectively. Estimate the maximum sampling interval required for the signal given by $Y[t] = X_1[t]\ X_2[t]$. [2] (**70 Ch**)

6.4.2 Sampling of band-pass signals and sub-sampling theory

1. Briefly explain the term sub-sampling theory. [2] (**74 Ch**, 72 Ka)

2. A signal x(t) = sinc($5\pi t$) is sampled (using uniformly spaced impulses) at a rate of 10 Hz, [2+2+1] (72 Ka)
   a. Sketch the sampled of signal (not to scale);
   b. Sketch the spectrum of the sampled signal for the range $|f|$ ¡ 30 Hz
   c. Explain whether you can recover the signal x(t) from the sampled signal.

6.4.3 Non-ideal sampling pulses (aperture effect)

1. Briefly explain the term aperture effect. [2] (**74 Ch**)

6.4.4 Non-ideal reconstruction filter

6.4.5 Time-limitedness of the signal (aliasing effects)

1. Define the aliasing and aperture effects. [2] (71 Shr) [3] (**73 Ch**) [4] (**75 Ch**) [5] (73 Shr)
   $|\rightarrow$ with remedy solutions. [6] (74 Ash)

2. What is aliasing and how can it be minimized? [3] (70 Asa)

7 Pulse Modulation Systems

(6 Hours / 6 Marks)

7.1 Analog Pulse Modulation

7.1.1 Pulse Amplitude Modulation (PAM): generation, bandwidth, spectrum, reconstruction

1. Write short notes on: PAM, PWM and PPM. [5] (**79 Bh**)
2. Define PMA, PWM and PPM with corresponding waveforms. [1.5] (**70 Ch**)

7.1.2 Pulse position and pulse width modulations: generation, bandwidth requirements

1. Differentiate between pulse amplitude and pulse position modulation. [3] (**81 Bh**)
2. Explain the generation of pulse width modulation. [4] (**81 Bh**)

7.2 Pulse Code Modulation (PCM)

7.2.1 PCM as a result of analog to digital conversion

1. Define PCM. [2] (81 Ba)
2. Explain working principle of PCM with necessary figures and equations. [5] (73 Shr)
3. A PCM system uses a uniform quantizer followed by a 7 bit binary encoder. The bit rate of the system is equal to $50 \cdot 10^6$ bits/sec. What is the maximum message signal bandwidth for which the system operates satisfactorily? [3] (73 Shr)
4. What are the design goals of digital modulation techniques? [2] (**73 Ch**)

7.2.2 Uniform quantization

1. Define quantization. [2] (**80 Bh**)

7.2.3 Quantization noise

7.2.4 Signal to quantization noise ratio (SQNR) in uniform quantization

1. Find the SQNR of PCM. [5] (**79 Ch**)
2. Derive the expression for the SQNR of uniformly quantized PCM. [4] (**72 Ch**,**71 Ch**) [6] (*71 Ma*)
3. Derive expression for evaluating SQNR for uniform quantization in terms of number of levels and number of bits per source symbol. [7] (74 Ash)
4. Find SQNR in uniform quantization in terms of no. of bits per source sample. [8] (70 Asa)
5. What is the relation between SQNR value and bit used for coding? [2] (*71 Ma*)
6. With necessary derivations show that in case of PCM, SQNR increases approximately by 6dB for each extra bit used. [6] (75 Ash)
7. Derive the expression for evaluating SQNR for linear quantization. [6] (**80 Bh**)

8. Determine the SQNR for delta modulation with no slope overload condition. [5] (**76 Ch**)

9. A television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Given that the number of quantization level is 512. Determine: [6] (**74 Ch**)
   a. Code word length
   b. Transmission bandwidth
   c. Bit rate
   d. SQNR

   |→For bandwidth = 4.8 MHz, other values same. [1+1.5+1.5+1.5] (**70 Ch**)

7.3 Advanced Quantization Techniques

7.3.1 Non-uniform quantization

1. Elaborate on the technique of implementing non-uniform quantization using uniform quantization technique. [5] (**81 Bh**)

2. Differentiate between uniform quantization and non-uniform quantization. [2] (74 Ash) [4] (**73 Ch**)

3. Find non-uniform quantization done for speech signal? [2] (75 Ash)

4. Explain basic process of Non-uniform quantization including companding technique of its realization. [3] (72 Ka)

5. Why is non uniform quantization required? [2] (**74 Ch**) [3] (**76 Ch**)
   |→for speech signal. [2] (73 Shr)

6. Explain any one algorithm for non-uniform quantization. [3] (**74 Ch**)

7.3.2 Improvement in average SQNR for high crest factor signals

7.3.3 Companding techniques (u and A law companding)

1. Explain generation of pulse width modulation. [3] (**81 Bh**)

2. What is companding? [1] (**79 Bh**, 76 Ash) [2] (**75 Ch**, 71 Shr)

3. Why is companding necessary? [1] (76 Ash)

4. Explain any two types of companding techniques. [5] (76 Ash)

5. Explain about companding laws. [2] (75 Ash)

7.3.4 Data rate and bandwidth of a PCM signal

7.4 Differential PCM and Delta Modulation

7.4.1 Differential PCM: encoder, decoder

1. Illustrate, the DPCM scheme that overcomes the disadvantages of PCM. [3] (**78 Ch**)

2. Explain DPCM. [2.5] (80 Ba) [4] (**70 Ch**)
   |→Briefly explain DPCM along with derivation and block diagram. [4] (**75 Ch**)
   |→Explain the working principle of DPCM with necessary transmitter and receiver. [4] (75 Ash)

3. Explain why DPCM is preferred over PCM? [2] (75 Ash)
|→Why is DPCM superior over PCM? [2] (**71 Ch**)

4. Explain DPCM working principle with necessary figures and equations. [4] (**71 Ch**)

7.4.2 Delta Modulation: encoder, decoder

1. Explain the Delta and Adoptive Delta Modulations encoders and decoders with their derivations and diagram. [6] (**80 Ch**)

2. Explain the Delta Modulation encoder and decoder with its derivations and diagram. [6] (**75 Bh**)

3. Compare DPCM and Delta Modulation. [2.5] (80 Ba)

4. List out merits and demerits of Delta and Adoptive Delta Modulations. [2] (**80 Ch**)
|→Explain the principles of Adaptive Delta Modulation (ADM). [5] (**79 Bh**)

5. Write short notes on: Adaptive Delta Modulation. [4] (72 Ka)

6. Compare PCM, DPCM and Delta Modulation. [3] (**79 Ch**)
|→Compare between PCM and DM. [1] (**75 Bh**)

7. A delta modulator system is designed to operate at 5 times the Nyquist rate for a signal having a bandwidth equal to 3kHz bandwidth. Calculate the maximum aplitude of a 2kHz sinusoidal for which the delta modulator does not have slope overload. The given step size is 250 mV. [4] (**78 Ch**)

8. Find the Nyquist rate and the interval for m(t) = $\sin^2(400\pi t)$ [4] (76 Ash)

9. Find the Nyquist rate and the interval for $dfrac12\pi \cos(400\pi t) \cos(1000\pi t)$. [4] (75 Ash)

10. A delta modulator is used to encode speech signal band-limited to 5 kHz with a sampling frequency of 200 kHz. For maximum signal amplitude of $A_{max} = 1$, find:
    a. Minimum step size to avoid slope overloading.
    b. Assuming the speech signal to be sinusoidal, find signal to quantization noise ratio.
    c. Determine the minimum transmission bandwidth.

11. The bandwidth of a TV plus audio signal is 4.5 MHz. If this signal is converted to PCM with 1024 quantization levels, determine bit rate of the resulting PCM signal. Let us assume that the signal is sampled at a rate 20% above the Nyquist rate. [4] (**75 Ch**)

12. A signal g(t) = $10\cos(20\pi t)\ \cos(200\pi t)$ is sampled at the rate of 250 samples per second, then (i) determine the spectrum of the resulting sampled signal, (ii) specify the cut-off frequency of the ideal reconstruction filter so as to recover g(t) from its sampled version, and (iii) determine the Nyquist rate for g(t). [3] (**73 Ch**)

13. The information in an analog waveform with maximum frequency 4 kHz is ot be transmitted over a 16-level PCM system. [6] (**73 Ch**)
    a. What would be the maximum number of bits per sample?
    b. What is the minimum sampling rate and bit rate?

14. A delta modulator is used to encode speech signal band-limited to 3 kHz with sampling frequency 10 kHz. For maximum signal amplitude of $A_{max} = 1$, find: [2+3+1] (72 Ka)
    a. Minimum step size to avoid slope overloading.
    b. Assuming the speech signal to be sinusoidal, find SQNR
    c. Determine the minimum transmission bandwidth.

7.4.3 Noises in DM and SQNR

1. Derive the expression for SQNR in delta modulation. [6] (**70 Ch**, 73 Shr)
   $|\rightarrow$ for a single-tone analog signal. [7] (81 Ba)

7.4.4 Comparison between PCM and DM

7.5 Parametric Speech Coding

7.5.1 Vocoders

8 Baseband Data Communication Systems

(6 Hours / 7.5 Marks)

8.1 Information Theory Fundamentals

8.1.1 Information and Entropy

1. Define information and entropy. [2] (81 Ba, 70 Asa)
2. Differentiate between message and information? [2] (**70 Ch**)
3. Relate the message, the entropy and information. [6] (**80 Bh**)
   |→State and derive the relation between entropy and information rate. [4] (**76 Ch**)
4. A discrete source one of 5 possible symbols per 10μs statistically independent manner. The symbol probabilities are $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16},$ and $\frac{1}{16}$. Find (a) Symbol rate (b) Source entropy (c) Information rate. [3] (**70 Ch**)
5. A continuous signal is band limited to 5 kHz. The signal is quantized in 6 levels of a PCM system with the probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32},$ and $\frac{1}{32}$. Calculate the entropy and information rate. [5+5] (74 Ash)
6. A fixed length encoder generates six symbols with probabilities { 0.2, 0.15, 0.25, 0.05, 0.3, 0.05 } and encoded by unique code word. Find entropy and maximum code efficiency. [3] (72 Ka)
7. A discrete source one of 6 possible symbols per 10μs statistically independent manner. The symbol probabilities are $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16},$ and $\frac{1}{16}$ respectively. Calculate symbol rate, entropy and information rate. [6] (70 Asa)

8.1.2 Shannon Hartley Channel capacity theorem

1. Explain Shannon-Hartley channel capacity theorem. [1] (**71 Ch**) [2] (**72 Ch**)
   |→with example [3] (**80 Ch**)
   |→with implication and theoretical limits. [6] (80 Ba, 75 Ash)
   |→|→with example. [8] (76 Ash)
2. Write down theoretical limitations of Shannon-Hartley theorem. [3] (**72 Ch,71 Ch**)
3. Derive the expression which shows the limit of Shannon's channel capacity theorem when bandwidth tends to infinity. [6] (**76 Ch**)
4. Calculate the upper limit of the channel capacity as the bandwidth of the channel (B) tends to infinity. [4] (81 Ba, 70 Asa)
   |→Show that channel capacity 1.44S/No, when channel bandwidth is tends to infinity. [10] (80 Ba) [4] (75 Ash)
5. State and explain Nyquist Channel Capacity Thereom. [8] (**80 Bh**)

8.2 Line Coding

1. Write short notes on: Line codes. [4] (72 Ka)

8.3 Unipolar (NRZ, RZ)

1. Define
   |→Unipolar line code. [2] (**79 Ch**)
   |→Unipolar RZ line code. [2] (**79 Ch**)

2. For given bit sequence and requirement, draw waveform(s):

Bits	NRZ	RZ	Marks
110011000001	✓		2: **80 Ch**
1101010111		✓	2: **80 Ch**
10110000100000000111		✓	1.5: 76 Ash
10110101		✓	1: 75 Ash
1011000010000000001		✓	1.5: **74 Ch**

8.4 Polar

1. Define polar line code. [2] (**79 Ch**)

8.4.1 NRZ & RZ

1. For given bit sequence and requirement, draw waveform(s):

Bits	NRZ	RZ	Extra	Marks
100000000001101		✓	(a)	2.5: **81 Bh**
1101010111	✓			2: **79 Ch**
100111010		✓		2 **78 Ch**
1001101101		✓		1.5: **81 Bh**
1101010111	✓	✓		1: 81 Ba
1011001010	✓	✓		1: **70 Ch**, 73 Shr
1101010011	✓	✓		1: **73 Ch**
10110000100000000111	✓	✓		1.5: 76 Ash
10110101	✓			1: 75 Ash
1011000010000000001	✓			1.5: **74 Ch**
1001001101	✓	✓		1.5: **72 Ch**

2. Extra:
   (a) Compare bandwidths with other forms

8.4.2 Manchester & Differential Manchester

1. Define Manchester line code. [2] (**79 Ch**)

2. For given bit sequence and requirement, draw waveform(s):

Bits	Manchester	Diff. Manchester	Extra	Marks
100000000001101		✓	(a)	2.5: **81 Bh**
110011000001	✓			2: **80 Ch**
1101010111	✓			2: **79 Ch**
100111010	✓			2: **78 Ch**
1001101101	✓			1.5: **81 Bh**
1101010111	✓			1: 81 Ba
1011001010	✓			1: **70 Ch**, 73 Shr
1101010111	✓			1: **73 Ch**
10110000100000000111	✓			1.5: 76 Ash
10110101	✓			1: 75 Ash
1011000010000000001	✓			1.5: **74 Ch**
1001001101	✓			1.5: **72 Ch**

8.5 Bipolar

1. Define bipolar line code. [2] (**79 Ch**)

2. For given bit sequence and requirement, draw waveform(s):

Bits	NRZ	AMI	B8ZS	HDB3	Extra	Marks
100000000001101			✓	✓	(a)	2.5: **81 Bh**
110011000001				✓		2: **80 Ch**
1101010111	✓					2: **79 Ch**
100111010	✓	✓				2: **78 Ch**
1001101101	✓	✓				1.5 **81 Bh**
1101010111		✓				1: 81 Ba
1011001010		✓				1: **70 Ch**, 73 Shr
1101010011		✓				1: **73 Ch**
10110101		✓				1: 75 Ash
1011000010000000001		✓				1.5: **74 Ch**
1001001101		✓				1.5: **72 Ch**

8.5.1 Properties and comparisons of line codes

8.6 Baseband System Design

8.6.1 Baseband data communication systems

8.6.2 Inter-symbol interference (ISI)

1. Define ISI. [2] (**81 Bh**,**80 Bh**,**74 Ch**,**72 Ch**, 73 Shr, 72 Ka, 71 Shr) [6] (**75 Ch**)

2. Illustrate the methods to reduce Intersymbol Interference. [4] (**79 Bh**)

3. Explain ISI in baseband digital communication system with derivations. [4] (74 Ash)

4. Explain the ideal and practical solutions of ISI. [3+3] (74 Ash)

5. Explain the ideal and practical solutions for zero ISI. [6] (**72 Ch**)

6. State Nyquist Criteria for zero ISI in both time and frequency domain. [6] (**71 Ch**)

8.6.3 Pulse shaping (Nyquist, Raised-cosine) and bandwidth considerations

1. Elaborate Nyquist pulse shaping criterion for mitigation of ISI. [6] (**81 Bh**)
   |→What are the physical constraints in implementing Nyquist pulse shaping criteria? [4] (**80 Bh**)

2. State Nyquist conditions for zero ISI. [2] (**80 Bh**, 72 Ka, 71 Shr)
   |→State Nyquist pulse shaping criteria for zero ISI. [2] (70 Asa)

3. Discuss any one pulse shaping method of ISI reduction. [4] (70 Asa)

8.7 Advanced Signaling Techniques

8.7.1 Correlative coding techniques

1. Explain any one type of correlative coding technique with its impulse response and transfer function. [6] (**73 Ch**)

8.7.2 Duobinary and modified duocoders

1. Mention how can we minimize error in duo-binary coding. [2] (81 Ba, **73 Ch**)

2. Explain duo-binary encoding with example. [4] (72 Ka)

3. Explain modified Duobinary coding technique with precoder, illustrating with input sequence 01001101. [6] (**80 Bh**)
   |→input sequence 10110011. [4] (**75 Bh**)
   |→input sequence 0010110. [5] (73 Shr)

4. Explain duo-binary encoding with the use of precoder. [6] (**74 Ch**)

5. What are two major difficulties with duo binary encoding method and explain how can they be solved? [6] (**71 Ch**)

8.7.3 M-ary signaling and comparison with binary signaling

1. (Assumed) What are the significances of multilevel modulation? [2] (**78 Ch**)

2. Write short notes on: M-ary baseband communication system. [5] (**79 Bh**,**75 Bh**,**72 Ch**)

3. Write short notes on: M-ary baseband data communication. [5] (**75 Bh**)

8.8 System Performance Analysis

8.8.1 The eye diagram

1. Write short notes on: The eye diagram. [4] (**81 Bh**) [5] (**76 Ch**,**72 Ch**)
   |→on: eye diagram and its significance. [4] (72 Ka)

9 Bandpass (Modulated) Data Communication Systems

(4 Hours / 5 Marks)

9.1 Binary Digital Modulations

9.1.1 ASK, PSK, FSK, DPSK, QPSK implementation

1. Explain the modulation scheme for MSK with appropriate example. [4] (**80 Ch**)
2. Discuss Gaussian Minimum Shift Keying (GMSK) with its advantage. [5] (81 Ba)
3. Briefly explain generation and detection of Phase Shift Keying (PSK) with necessary illustration. [5] (81 Ba)
4. What do you understand by differential coding? Explain differential phase shihft keying modulation and detection with example and diagrams. [5] (71 Shr)
5. Explain coherent binary PSK modulation technique with its signal space diagram, modulator and demodulator. [8] (**73 Ch**)
6. Explain BSPSK modulation technique with its relevant diagram and signal space diagram. [5] (**78 Ch**)
7. Why DPSK is preferred than PSK? [1] (72 Ka)
8. What is DPSK? [1] (70 Asa)
9. How it can DPSK be implemented? [3] (70 Asa)
10. Explain the modulator, demodulator and signal space diagram FSK modulation. [6] (**72 Ch**)
    |→For QPSK with derivation. [8] (**70 Ch**)
11. Explain the generation and non coherent detection for BFSK signal and also the signal space diagram for a BFSK signal. [8] (**76 Ch**)
12. Explain the modulator, demodulator, and signal space diagram for QPSK modulation with relevant derivations. [10] (80 Ba)
    |→Explain the modulation and demodulation techniques used in QPSK. [8] (75 Ash)
13. Explain QPSK with its transformation as well as receiver block diagram. [4] (**78 Ch**)
14. Explain the modulator, demodulator and signal space diagram for DPSK system. [2+2+1] (72 Ka)

9.1.2 Properties and comparisons of binary modulations

9.2 M-ary Digital Modulations

9.2.1 Quadrature amplitude modulation (QAM) systems

1. What is QAM, why is it necessary? [3] (**74 Bh**)
   |→What is the significance of constellation diagram? [2] (**80 Ch**)
2. Explain generation and detection of QAM wave. [4] (**74 Bh**)
3. Explain the working of Quadrature Amplitude Modulation (QAM) with appropriate modulation, demodulation block, and constellation diagram. [5] (**81 Bh**)
4. Define an efficient constellation diagram of a 32-QAM. [3] (**81 Ch**)

9.2.2 Four phase PSK systems

9.3 Demodulation and Applications

9.3.1 Demodulation of binary signals (coherent and non-coherent)

1. (Assumed) Derive the expression for figure of merit for the SSB receiver. [5] (**81 Bh**)

2. Write short notes on: Coherent detector. [4] (72 Ma)

9.3.2 Modems and its applications

1. What is modem? Discuss the modes of operation of modems. [4] (70 Asa)

9.4 Numerical

1. A band-limited m(t) with a bandwidth of 3 kHz is sampled at a rate equal to twice the Nyquist rate. The step size used during the quantization process is 0.5% of the peak amplitude m_p. Assuming binary encoding, determine the minimum bandwidth of the channel to transmit the encoded binary signal. [5] (81 Ba)

10 Random Signals and Noise in Communication Systems

(6 Hours / 7.5 Marks)

10.1 Random Processes

10.1.1 Random variables and processes

1. What do you mean by random process? [2] (72 Ka, 71 Shr)

10.1.2 Statistical and time averaged moments

1. Define moment and central moment of continuous random variable. Show that first central moment is always zero. [2] (**71 Ch**)

10.1.3 Stationary and ergodic processes

1. What is Ergodic Stcastic Process? [2] (**74 Ch**, 81 Ba, 76 Ash)

10.1.4 PSDF and AC function of an ergodic random process

10.2 Noise Characterization

10.2.1 White noise and thermal noise

10.2.2 Band-limited white noise

10.2.3 PSDF and AC function of white noise

10.3 Noise in Linear Systems

10.3.1 Passage of random signals through a LTI system

1. (Assumed) Show that the output signal of LTI system is also a wide sense stationary process if the input to it also a WSSP. [6] (81 Ba)

2. (Assumed) Find mean and AC function at the output when a WSSP signal is passed through LTI system. [5] (70 Asa)

3. (Assumed) Explain with necessary derivation passage of wide-sense random signals through an LTI. [10] (76 Ash)
   |→Explain with necessary derivation passage of wide-sense random signals through an LTI. [10] (**74 Ch**)

4. Show that the output of LTI is WSSP if the input is also a WSSP. [6] (71 Shr)

10.3.2 Ideal low-pass and RC filtering of white noise

1. Explain and compare ideal and practical RC filtering of white noise with respect to change in auto-correlation function. [8] (**75 Ch**)

2. Explain the approximation of the matched filter for a rectangular pulse using a single pole RC low pass filter with variable bandwidth. [6] (72 Ka)

3. Explain white noise with its PSDF and autocorrelation function. [4] (72 Ka)

4. Determine the noise equivalent bandwidth of RC-LPF and that of ideal LPF of zero frequency response one. Also, find output noise power of this RC-LPF when input is white noise. [4] (**71 Ch**)

10.3.3 Noise equivalent bandwidth of a filter

1. Explain Noise Equivalent Bandwidth represents of a filter. [4] (**81 Bh**)
   |→Write short notes on: Noise Equivalent Bandwidth. [5] (**72 Ch**)

2. Define Noise Equivalent bandwidth. [2] (**73 Ch**) [3] (70 Asa) [4] (80 Ba)

3. Find noise equivalent bandwidth of the first order RC low pass filter. [6] (80 Ba)

10.4 Optimum Detection and Matched Filter

10.4.1 Optimum detection of a pulse in additive white noise

1. Define optimum detector. [1] (**71 Ch**) [2] (**80 Ch, 79 Ch**,**74 Ch**, 76 Ash)
   |→Define optimum detection. [2] (**72 Ch**)

2. Define matched filter. [2] (72 Ka) [5] (**79 Bh**)

3. Realize matched filter with relevant mathematical support. [4] (70 Asa)

4. Find the impulse response of optimum detector in the presence of additive white noise. [5] (**79 Bh**) [6] (**80 Ch**,**81 Bh**,**71 Ch**, 76 Ash)

5. Derive the expression of impulse response of the matched filter. [6] (81 Ba) [8] (74 Ash)

6. What are the physical constraints in implementing Nyquist pulse shaping criteria? [4] (**80 Bh**)

10.4.2 The matched filter concept and realization (time correlators)

10.4.3 Matched filter for a rectangular pulse

10.4.4 Ideal LPF and RC filters as matched filters

1. Show that the impulse response of the matched filter is reverse delayed version of the input signal. [6] (**79 Ch**,**74 Ch**, 71 Shr) [8] (73 Shr) [10] (**73 Ch**)

2. Show that the impulse response of the optimum detector network is the time shifted replica of the incoming signal. [5] (**72 Ch**)

3. Explain the approximation of the matched filter for a rectangular pulse using an Ideal low pass filter with variable bandwidth. [6] (**70 Ch**)

10.5 Error Performance in Baseband Systems

10.5.1 Performance limitation due to noise

10.5.2 Error probabilities in binary and M-ary baseband systems

1. (Assumed) Explain communication impairments with examples. [2] (**79 Ch**)

2. Evaluate the error probability in a binary communication system with appropriate expression. [7] (**81 Bh**,**70 Ch**) [8] (**80 Bh**)

3. Derive the expression for evaluating error probability in binary baseband system and compare with M-ary system. [8] (76 Ash)

4. Derive the expression for evaluating error probability in binary communication system. [7] (**79 Bh**)

5. Derive expression for evaluating error probability of M-ary system. [5] (71 Shr) [8] (75 Ash)

11 Noise performance of Bandpass Communication Systems

(6 Hours / 8.5 Marks)

11.1 Noise in AM Systems

11.1.1 Effect of noise in envelop and synchronous demodulation of DSB-FC AM

1. Evaluate the noise performance of the coherent detection of DSB-FC using necessary derivations. Use the result ot evaluate the noise performance of the signal-tone DSB-FC modulation with 100% modulation. [2] (81 Ba)
2. With necessary derivation, compare noise performance of DSB-AM, DSB-SC, SSB-SC. [8] (**72 Ch**)
3. With necessary derivations, compare the noise performance of DSB-SC and SSB-SC modulations in analog communication system. [6] (71 Shr)

11.1.2 Expression for gain parameter (output SNR to input SNR)

1. Derive the expression for evaluation the gain parameter ($\mathrm{SNR}_0/\mathrm{SNR}_1$) of non-coherent FM detector. [8] (**71 Ch**, 73 Shr)

11.1.3 Threshold effect in nonlinear demodulation of AM

11.1.4 Gain parameter for synchronous demodulation of DSB-SC and SSB

1. Calculate the gain parameter in DSB-FC with envelop detection. [5] (70 Asa)

11.2 Noise in FM Systems

11.2.1 Effect of noise (gain parameter) for non-coherent demodulation of FM

11.2.2 Threshold effect in FM

1. Write short notes on: Threshold effect in demodulation of FM. [4] (*71 Ma*)
2. What is threshold effect in FM? [7] (**79 Bh**)
   |→and how it can be minimized? [3] (**70 Ch**)
3. Explain threshold effect in detection of FM signal. [4] (**76 Ch**)
4. What are the techniques for the mitigation of threshold effect in FM? [3] (**79 Bh**)
5. Compute the figure of merit of non coherent FM system and explain the threshold effects. [6+2] (74 Ash)
6. With necessary derivations, explain the threshold effect in envelope detector for DSB-FC modulation in analog communication system. [7] (72 Ka)

11.2.3 Use of pre-emphasis and de-emphasis circuits in FM

1. Why do we need pre-emphasis and de-emphasis circuits in FM? Explain. [4] (**80 Ch**)
   |→Why pre-emphasis and de-emphasis circuits are required in commercial FM broadcasting? [2] (70 Ma)

2. Why pre-emphasis and de-emphasis circuits are used in commercial FM broadcasting? [3] (**75 Bh**)

3. Write short notes on: Pre-emphases and de-emphases. [4] (**72 Ash**)

4. Why pre-emphases is needed in FM during transmission? [2] (71 Ma)

11.3 Comparison of Analog Modulation Schemes

11.3.1 Comparison of AM (DSB-FC, DSB-SC, SSB) and FM (Narrow and wide bands)

11.3.2 Comparison metrics: power efficiency, channel bandwidth and complexity

1. Compare A.M and F.M in terms of power efficiency and system complexity. [5] (80 Ba)
   |→Compare AM and FM in terms of power efficiency, brand width efficiency and system complexity. [3] (70 Asa)

2. Compare binary and M-ary scheme in terms of bandwidth efficiency and system complexity. [3] (71 Shr)

11.4 Noise in Digital Modulated Systems

11.4.1 Noise performance of modulated digital systems

11.4.2 Error probabilities for ASK, PSK, FSK, DPSK (coherent and non-coherent)

1. Derive the bit error probability for the coherent binary FSK system. [6] (81 Ba)

2. Calculate the error probability of coherent ASK. [4] (70 Asa) [5] (80 Ba)
   |→Derive the expression of error probability for coherent detection of ASK. [6] (74 Ash)
   |→Derive the expression for evaluating error probability for binary ASK system. [6] (**71 Ch**)
   |→Derive the expression for evaluating error probability for binary ASK system and extend it to M-ary. [8] (73 Shr)

3. Find the error probability in coherent ASK and PSK detections and show that ASK requires double the average signal power than PSK for same error probability. [3] (**72 Ch**)

4. Derive the expression of error probability for coherent detection of Phase Shift Keying (PSK). [6] (**76 Ch**)

5. Derive the expression of error probability for binary PAM signal. [5] (72 Ka)

11.4.3 Comparison of digital systems: bandwidth efficiency, power efficiency and complexity

12 Multiplexing

(2 Hours / 3 Marks)

12.1 Frequency Division Multiplexing (FDM)

12.1.1 Principle of FDM

1. Define FDM. [2] (**75 Bh**, 75 Ba) [4] (**79 Ch**)

2. Write short notes on: FDM. [5] (**73 Bh**, 70 Ma)
   |→Describe the principle of frequency division multiplexing (FDM). [2] (**71 Bh**)

12.1.2 FDM in telephony and hierarchy

1. Explain FDM hierarchy used in telephony. [5] (75 Ba) [6] (**79 Ch**,**75 Bh**)
   |→Write short notes on: FDM in telephony. [4] (**76 Bh**, 76 Ba)
   |→Write short notes on: FDM Telephone Hierarchy. [5] (**70 Bh**)

12.1.3 Frequency Division Multiple Access (FDMA) systems (SCPC, DAMA, SPADE)

1. What do you understand by FDMA? [2] (72 Ma, 71 Ma)

2. Briefly explain SCPC and Dama types of FDMA. [4] (**71 Bh**)

12.1.4 Filter and oscillator requirements in FDM

1. Write short notes on: Filter and oscillator requirements in FDM. [5] (**80 Ch**,**73 Bh**, 73 Ma)

2. Write short ntoes on: FDMA in satellite communications. [4] (**74 Bh**) [5] (**73 Bh**)

3. Write short notes on: Satellite communication system. [4] (**72 Ash**)

12.2 Time Division Multiplexing (TDM)

1. Write short notes on: FDM vs TDM. [5] (**81 Ch**)
   |→Compare TDM and FDM. [3] (**79 Ch**)
   |→Compare FDMA and TDMA. [3] (**78 Ch**)

2. *(Attached with Qno.1 of (**79 Ch**), but it has no description whatsoever, so i've put it here.)*
   Show that for voice application. [2] (**79 Ch**)

3. An audio signal of frequency 4 kHz and maximum dynamic range of ± 2.4V is digitized by PCM system with its bit rate of 64 kHz. Calculate numbers of bits per sample, quantization noise and $SQNR_{dB}$. Estimate the minimum bandwidth required for TDM of 10 such audio signals (assume no extra framing and synchronization bits). [3+2] (72 Ka)

12.2.1 Time Division Multiplexing with PCM

12.2.2 Data rate and bandwidth of a PCM signal

1. Prove that signaling rate for time division multiplexing of 24 voice channels is equal to 1.544 Mbps. [8] (*71 Ma*)

12.2.3 The T1 and E1 TDM PCM telephone hierarchy

1. Write short notes on: T1 and E1 hierarchy used in telephony. [5] (**81 Ch**)
2. Draw T1 and E1 telephony hierarchy. [2] (**78 Ch**)
3. Explain T1 digital carrier system with its multiplexing hierarchy. [5] (**79 Bh**)
4. Explain the differences between T1 and E1 digital hierarchy. [4] (75 Ash)
5. What are the signaling (bit) rate and bandwidth requirement for the T1 and E1 digital carrier systems? [3] (**70 Ch**)
6. Explain T1 hierarchy of TDM-PCM telephony. [4] (71 Shr)
7. Explain E1 TDM-PCM Telephone Hierarchy. [4] (**74 Ch**) [5] (80 Ba)
8. Describe E1 fram and its TDM hierarchy along with signaling rate. [3+3] (74 Ash)
9. Compare E1 and T1 TDM PCM hierarchy. [4] (**80 Ch**) [5] (**79 Ch**)
10. Write short notes on: The E1 digital hierarchy. [5] (**76 Ch**)
11. Explain E1 digital hierarchy as related to telephony system. [4] (**72 Ch**,**71 Ch**)
12. Explain the use of FDM in telephony hierarchy. [6] (72 Ma, 71 Ma)
13. With the help of the frame diagrams, discuss T1 and E1 hierarchy of TDM-PCM telephony. [8] (**80 Bh**)

13 Error control coding techniques

(6 Hours / 8.5 Marks)

13.1 Fundamentals of Error Control

13.1.1 Basic principles and types of error control coding

1. Differentiate error-detection and error-correction. [2] (**79 Ch**)

2. Write short notes on: Error detection and correction capability of a codeword. [4] (**81 Bh**)

13.1.2 Basic definitions: hamming weight, hamming distance, minimum weight

1. Define Hamming Weight and Hamming Distance.
[2] (**75 Ch**,**74 Ch**,**72 Ch**,**71 Ch**, 74 Ash, 70 Asa) [3] (73 Shr) [4] (76 Ash)

2. What is the importance of hamming distance and hamming weight? [2] (71 Shr)

13.1.3 Hamming distance and error control capabilities

1. What is the significance of (d_{min}) minimum hamming distance? [2] (**81 Bh**)

13.2 Parity Block Codes

1. Explain with example, the syndrome decoding method in linear block coding. [5] (71 Shr)

2. Prove that (4, 3) even parity code is a linear block code and (4, 3) odd parity code is not a linear code. [7] (81 Ba)

3. For a (6, 3) code, the parity check matrix is given by H = $\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$. Determine whether a received vector 100101 is erroneous. [5] (**75 Ch**)

4. Validate the code if received code vector code is [100011] given by that H = $\begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$

5. A (6, 3) error control code has the following parity check matrix. [2+4+4] (**81 Ch**)
$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$
a. Determine the generator matrix G.
b. Generate a codeword for message 110.
c. Check whether the received code word 1101111 has error or not.

6. For a code vector x = (0111000), and the parity check matrix H given below. Prove that, the given code is valid. [4] (**72 Ch**,**71 Ch**)
$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}_{3\times7}$$

13.3 Binary Cyclic Codes

1. What is binary Cyclic code? [1] (70 Asa) [2] (80 Ba)

2. Find systematic and non-systematic codes for (7, 4).

Generator	Message	Year
$x^3 + x^2$	1011	4: 70 Asa, 6: 74 Ash, 8: 80 Ba
$1 + x + x^3$	1011	6: **70 Ch**
$p^3 + p + 1$	0101	8: 75 Ash
$1 + x + x^3$	1011	5: **78 Ch**
$1 + x + x^3$	1010	6: **73 Ch**
$1 + x + x^3$	1101	6: **81 Bh**
$x^3 + x^2 + 1$	1011	5: 76 Ash

13.4 Convolutional Codes

13.4.1 Implementation of convolutional codes

1. Why convolution coder is better than block coder? [2] (**78 Ch**,**73 Ch**)

2. Explain convolution coder with suitable example. [6] (72 Ka)

3. Write short notes on: Convolution Codes. [4] (*71 Ma*)
   |→Write short notes on: Convolution Coding. [4] (**80 Bh**)
   |→Write short notes on: Convolution Coder. [5] (**79 Bh**)
   |→Write short notes on: Convolution Coder with example. [5] (**75 Bh**)

4. Explain the operation of a 1/3 convolution encoder. [6] (**74 Ch**)

13.4.2 Code tree and trellis representation

1. Write short notes on: Trellis diagram of a message 1010 using $\frac{1}{2}$ rate three-state memory block convolutional encoder. [5] (**81 Ch**)

2. Determine the encoded sequence for the following input message. [8] (**80 Ch**)
   (m_0, m_1, m_2, m_3, m_4) = (1 0 0 1 1) using convolutional encoder having shift registers with three flip flops. Draw state and trellis diagrams. [8] (**80 Ch**)

13.4.3 Decoding algorithms for convolutional codes

1. Design a convolution encoder having code-rate of $\frac{1}{2}$. Also, draw the code-tree and trellis diagram for the same assuming any three-bit input. [4+4] (**79 Ch**)

2. The 1/3 rate convolutional encoder with constraint length equal to 3 has following three generator sequences each of length 3. [3] (**76 Ch**)
   $\left(g_0^{(1)},\ g_1^{(1)},\ g_2^{(1)}\right) = (0, 1, 1)$
   $\left(g_0^{(2)},\ g_1^{(2)},\ g_2^{(2)}\right) = (1, 0, 1)$
   $\left(g_0^{(3)},\ g_1^{(3)},\ g_2^{(3)}\right) = (1, 1, 0)$
   Determine the encoded sequence for the following input message (m_0, m_1, m_2, m_3, m_4) = (1 0 0 1 1)

14 I don't know where these fit

1. Write short notes on: linear prediction theory. [4] (**81 Bh**)
   $|\rightarrow$Write short notes on: linear prediction theory/coding [6] (**70 Ch**)

2. Write down the significant of variable length coding with relevant example. [2] (72 Ka)

3. What is capture effect? [2] (70 Asa)